

APM1110_FA5_Eduvala_Mikhael

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General Instructions

1. Read each problem carefully and write clear, step-by-step solutions for all items.
2. Show all complete calculations, formulas used, and final answers. Round off final numerical values to four decimal places unless specified otherwise.
3. Format your completed work into a single PDF file or a well-documented Markdown/LaTeX file.
4. Create a designated public repository on your GitHub account named APM1110-FA5.
5. Upload your file(s) to the repository main branch. Ensure all rendered images/formulas display correctly in your repository's README.md file or committed PDF.
6. Submit the public URL link of your GitHub repository to the course LMS portal before

1. Discrete Random Variables

A discrete random variable X has the following probability mass function (PMF):

$$P(X = x) = \begin{cases} k(2x + 1), & x = 1, 2, 3, 4 \\ 0, & \text{otherwise} \end{cases}$$

- (a) Find the value of the constant k . For a probability mass function, everything must add up to 1.

$$k(2(1) + 1) + k(2(2) + 1) + k(2(3) + 1) + k(2(4) + 1) = 1$$

$$k(3 + 5 + 7 + 9) = 1$$

$$24k = 1$$

Therefore, the value of the constant k is:

$$k = 1/24$$

(b) Compute the expected value $E[X]$ and the variance $\text{Var}(X)$. The expected value of a discrete random variable is

$$E[X] = \sum xP(X = x)$$

Since,

$$P(X = x) = k(2x + 1)$$

Then,

$$E[X] = \sum_{x=1}^4 k(2x + 1)$$

Substitute $k = 1/24$,

$$E[X] = 1/24[1(3) + 2(5) + 3(7) + 4(9)]$$

$$E[X] = \frac{3 + 10 + 21 + 36}{24}$$

$$E[X] = \frac{70}{24}$$

Therefore, the expected value is:

$$E[X] = 2.9167$$

Now, For the Variance $\text{Var}(X)$: Formula:

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

Calculate $E[X^2]$:

$$E[X^2] = \frac{1}{24}[1^2(3) + 2^2(5) + 3^2(7) + 4^2(9)]$$

$$E[X^2] = \frac{3+20+63+144}{24}$$

$$E[X^2] = \frac{230}{24}$$

$$E[X^2] = 9.5833$$

Then,

$$\text{Var}(X) = (9.5833) - (2.9167)^2$$

$$\text{Var}(X) = 9.5833 - 8.5071 = 1.0764$$

Therefore, $(b) E[X] = 2.9167$ and $\text{Var}(X) = 1.0764$

(c) Calculate: $P(X \geq 3)$ We need the probabilities for $X = 3$ and $X = 4$.

$$P(X \geq 3) = P(X = 3) + P(X = 4)$$

$$= k(2(3) + 1) + k(2(4) + 1)$$

$$= k(7 + 9)$$

$$= 16k$$

Since $k = \frac{1}{24}$,

$$P(X \geq 3) = \frac{16}{24}$$

$$P(X \geq 3) = 0.6667$$

2. Binomial Distribution

An IT system detects that roughly 8% of incoming external requests are malicious. In a random sample of 20 requests: Binomial Distribution should be used as each request can either be malicious or not malicious.

$$X \sim \text{Binomial}(n = 20, p = 0.08)$$

$$n = 20, \quad p = 0.08, \quad 1 - p = 0.92$$

Binomial Formula:

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$$

(a) What is the probability that exactly 2 requests are malicious? We need $P(X = 2)$

Using the formula, $P(X = 2) = \binom{20}{2} (0.08)^2 (0.92)^{18}$

$$P(X = 2) = 0.2711$$

(b) What is the probability that at least 3 requests are malicious? We need $P(X = 3)$

Use a complement, $P(X \geq 3) = 1 - P(X \leq 2)$

$$P(X \geq 3) = 1 - [P(X = 0) + P(X = 1) + P(X = 2)]$$

using the binomial distribution:

$$P(X \geq 3) = 1 - P(X \leq 2)$$

$$P(X = 3) = 0.2121$$

(c) Determine the mean and standard deviation of the number of malicious requests. For a binomial distribution,

$$\mu = np$$

Therefore,

$$\mu = (20)(0.08)$$

$$\mu = 1.6$$

The standard deviation is

$$\sigma = \sqrt{np(1 - p)}$$

Therefore,

$$\sigma = \sqrt{(20)(0.08)(0.92)}$$

$$\sigma = \sqrt{1.472}$$

$$\sigma = 1.2133$$

3. Poisson Distribution

A server receives an average of 4.5 web traffic hits per minute. Assuming the arrival rate follows a Poisson process: Poisson distribution:

For one minute,

$$\lambda = 4.5$$

The Poisson probability formula is

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

(a) Find the probability of receiving exactly 6 hits in a given minute. We need

$$P(X = 6)$$

Using the formula:

$$P(X = 6) = \frac{e^{-4.5} (4.5)^6}{6!}$$

$$P(X = 6) = 0.1281$$

(b) Find the probability of receiving at most 2 hits in a given minute.

$$P(X \leq 2)$$

$$P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2)$$

Then,

$$P(X \leq 2) = e^{-4.5} \left(1 + 4.5 + \frac{4.5^2}{2!} \right)$$

$$P(X \leq 2) = 0.1736$$

(c) Calculate the probability of receiving more than 10 hits in a 2-minute interval. There is an average of 4.5 hits per minute

Now, for a 2-minute interval, the new Poisson mean is:

$$\lambda = 4.5(2) = 9$$

Let Y represent the number of hits in 2 minutes.

Therefore,

$$Y \sim \text{Poisson}(9)$$

We need

$$P(Y > 10)$$

Using the complement,

$$P(Y > 10) = 1 - P(Y \leq 10)$$

$$P(Y > 10) = 0.2940$$

4. Continuous Probability & Normal Distribution

The processing time for a complex database query is normally distributed with a mean = 120 milliseconds and a standard deviation = 15 milliseconds. Therefore,

$$X \sim N(120, 15^2)$$

Conversion of a value to standard normal value:

$$z = \frac{x - \mu}{\sigma}$$

(a) What proportion of queries take longer than 135 milliseconds to process?

$$P(X > 135)$$

z-score:

$$z = \frac{135 - 120}{15}$$

$$z = \frac{15}{15} = 1$$

Therefore,

$$P(X > 135) = P(Z > 1)$$

$$P(Z > 1) = 1 - P(Z \leq 1)$$

$$P(Z > 1) = 0.1587$$

$$P(Z > 1) = P(X > 135) = 0.1587$$

Therefore, approximately 15.87% of the queries take longer than 135 milliseconds.

(b) Find the probability that a randomly chosen query takes between 100 and 130 milliseconds.

$$P(100 < X < 130)$$

Get the z scores:

z -score (100 milliseconds):

$$z_1 = \frac{100 - 120}{15}$$

$$z_1 = -1.3333$$

z -score (130 milliseconds):

$$z_2 = \frac{130 - 120}{15}$$

$$z_2 = 0.6667$$

Now that we got the two z -scores,

$$P(100 < X < 130) = P(-1.3333 < Z < 0.6667)$$

This can be calculated as

$$P(X < 130) - P(X < 100)$$

$$P(100 < X < 130) = 0.6563$$

(c) Find the 95th percentile of query processing times. The 95th percentile is the value x s.t.:

$$P(X \leq x) = 0.95$$

For a standard normal distribution,

$$z = 1.6449$$

We use

$$x = \mu + z\sigma$$

Therefore,

$$x = 120 + (1.6449)(15)$$

$$x = 144.6728$$

FINAL ANSWERS:

1. Discrete Random Variables

(a) $k = \boxed{0.0417}$

(b) $E[X] = \boxed{0.0417}$

(c) $Var(X) = \boxed{1.0764}$

(d) $P(X \geq 3) = \boxed{0.6667}$

2. Binomial Distribution

(a) $P(X = 2) = \boxed{0.2711}$

(b) $P(X \geq 3) = \boxed{0.2121}$

(c) $Mean = \boxed{1.6000}$

(d) $Standard\ Deviation = \boxed{1.2133}$

3. Poisson Distribution

(a) $P(X = 6) = \boxed{0.1281}$

(b) $P(X \leq 2) = \boxed{0.1736}$

(c) $P(X > 10) = \boxed{0.2940}$

4. Continuous Probability and Normal Distribution

(a) $P(X > 135) = \boxed{0.1587}$

(b) $P(100 < X < 130) = \boxed{0.6563}$

(c) $95th\ percentile = \boxed{144.6728\ ms}$